

Code Quality & Reusability Assessment

Date	06 July 2026
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Team

Code Layout Checklist

S.No	Code Parameter	Quality	Description	Followed (Yes/No)	Remarks
1	Consistent Indentation		Uniform spacing/tabs used throughout	Yes	4 spaces indentation standard in Python
2	Proper File Structure		Logical organization of source files	Yes	Clear split of train, app, static, templates
3	Meaningful Variables		Reflects variable purpose clearly	Yes	e.g. coapplicant_income_log, features_scaled
4	Method Names		Functions are descriptively named	Yes	e.g. preprocess_input(), load_model()
5	Code Comments		Inline comments explaining complex logic	Yes	Detailed math comments on risk flags and SMOTE

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6	Modular Design		Split into reusable functions/modules	Yes	Preprocessing isolated in helper functions
7	No Redundant Code		Duplicate or dead code removed	Yes	Clean code structures, unused model trials removed
8	Error Handling		Exceptions handled gracefully	Yes	Checks for missing rdf.pkl, invalid CSV schema

Reusable Components

S.No	Component Name	Language/Tech	Where Reused	Reusability Level
1	preprocess_input()	Python	Both app.py and train.py for preprocessing	High
2	StandardScaler asset	scikit-learn	Fit in training, transformed during live scoring	High
3	XGBoost classifier	XGBoost	Serialized and loaded on flask startup	High